

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Write an algorithm and a C-Program to obtain roots of non-linear equation using Newton Raphson Method.
2. Solve the following ordinary differential equation using shooting method. $(y'' + xy' - xy = 2x)$, with boundary conditions $(y(0)=1)$; and $(y(2)=10)$
3. Compare and contrast between Jacobi iterative methods and Gauss Seidal method? Solve the following equation using Gauss Seidal method. $(x + 2y + 3z = 5)$, $(2x + 8y + 22z = 6)$ and $(3x + 22y + 82z = -10)$
4. Use secant method to estimate the root of the equation $(x^2 - 5x + 6 = 0)$, with initial estimate $(x_1 = 4)$ and $(x_2 = 2)$ (EPS=0.05).
5. Solve the double integration using Simpson's 1/3 rule. $(\int_2^{2.6} \int_4^{4.4} \frac{dx}{dy} dy)$
6. What are the sources of errors? Discuss various types of errors encounters in numerical computation.
7. Fit a second order polynomial to the data in the table below:
X12345F(x)26122030
8. Solve the following differential equation $(\frac{dy}{dx} = 3x + \frac{y^2}{2})$ with $(y(0) = 1)$ for $(x = 0.2)$ $(h = 0.1)$ using Euler's Method.
9. Why Numerical Integration is required? Compute the integral: $(I = \int_{-1}^1 e^x dx)$ using composite trapezoidal rule for $(n = 4)$.
10. Evaluate $(\frac{dy}{dx})$ at $(x = 5)$ using Newton's forward interpolation formula using the following table.
X13579y-1.2012.80119.60472.801302.80
11. Find the Eigen values and Eigen vectors of the Matrix: $[A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}]$
12. Solve the Poisson's equation $(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 2x^2y^2)$ over the square domain $(0 \leq x \leq 3)$ and $(0 \leq y \leq 3)$ with $(f=0)$ on the boundary and $(h = 1)$.